

MANIKANDAN K.B.

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Summary

Software Developer with strong foundation in **Core Java, Python, OOP, and Application Development**. Skilled in **software maintenance, troubleshooting, and root cause analysis** across full-stack and AI-based systems. Experienced in writing **clean, testable, and efficient code** and delivering **high-volume, low-latency applications** across all phases of the **Software Development Lifecycle (SDLC)**. Committed to building **scalable, high-availability software** and continuously evaluating technologies to drive **system performance and architectural improvements**.

Technical Skills

- **Programming:** Core Java, Python, C++
- **Frameworks & Tools:** Spring (Basics), Git
- **Web Technologies:** HTML, CSS, JavaScript, React (Basics)
- **Databases:** SQL, MongoDB
- **Core Knowledge:** OOP, Data Structures, Application Development, Code Reviews, Software Development Lifecycle (SDLC)
- **IT Skills:** Troubleshooting, Software Maintenance, Root Cause Analysis, Performance Optimization

Experience

AI Developer Intern | Tech Vaseegrah

Jan 2025 – Jul 2025

- Designed and developed a **high-volume, low-latency voice AI application** processing real-time NLP commands with sub-second response time under concurrent user load.
- Architected an end-to-end pipeline – **Voice Input → STT → LLM Classification → Action Execution** – ensuring **high availability** and consistent throughput.
- Wrote **clean, modular, unit-testable code** in Python and JavaScript; enforced coding standards through **code reviews** and iterative debugging cycles.
- Participated across all **SDLC phases**: requirement analysis, system design, development, testing, build, and software release of production components.
- Fine-tuned **Whisper STT** on domain data and integrated **LLM-based classification** for module routing; improved command accuracy to **94%** and reduced execution latency by **35%**.
- Implemented **application logging, uptime monitoring, and performance metrics** to support root cause analysis and optimize service reliability.
- **Technologies:** Python, NLP, LLMs, Whisper STT, React, FastAPI, Node.js, MongoDB.

Freelance Software Developer | Self-Employed (Remote)

Aug 2025 – Present

- Developed and deployed a **full-stack CRM automation system** handling high-volume appointment workflows via trigger-based email, SMS, and AI voice pipelines; reduced manual processing by **80%**.
- Designed **RESTful API integrations** and authored technical specs, API documentation, and release notes ensuring full compliance with client system specifications.
- Evaluated and recommended automation technologies based on performance, scalability, and maintainability criteria for architectural review.

Technical Projects

End-to-End Voice Commerce Platform (Python, Whisper STT, LLM, React, FastAPI, MongoDB)

2025

- Built a **high-volume, low-latency voice-driven e-commerce application** supporting product search, cart management, and checkout via voice – zero manual UI dependency.
- Achieved **94% command accuracy** and **70% reduction in UI interaction time** through STT fine-tuning and LLM pipeline optimization.
- Designed **scalable backend APIs** (FastAPI, Node.js), integrated MongoDB for product and order management, and built a React frontend with real-time state updates.
- Implemented **structured logging, uptime dashboards, and latency tracking** for ongoing performance optimization and compliance with technical specifications.

Tomato Quality Classification System (Python, CNN, React)

2024

- Developed a **full-stack ML application** through complete SDLC: preprocessing, training, validation, REST API deployment, and React frontend.
- Achieved **97.5% binary** and **96.67% multiclass** accuracy using CNN with transfer learning; production-ready codebase with clean module separation and documented API endpoints.

College Complaint Management System (MongoDB, Express.js, React, Node.js)

2023

- Built a **role-based complaint tracking portal** with JWT authentication and role management for students, staff, and admin; delivered RESTful APIs and React UI.
- Improved complaint resolution efficiency by **40%** through automated status tracking and structured data models aligned to functional specifications.

Education

Master of Computer Applications (MCA) | SASTRA Deemed University

2023 – 2025

CGPA: 6.93

Bachelor of Computer Applications (BCA) | St. Joseph's College

2020 – 2023

CGPA: 7.80

Certifications

- C Programming for College Students – Udemy | AI Developer Internship Certification – Tech Vaseegrah